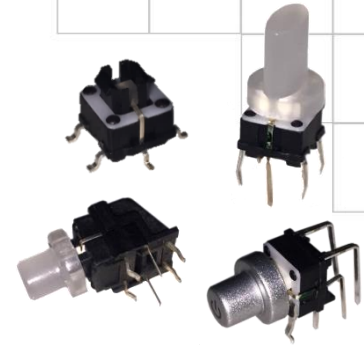




Series 228A

7.2 x 6.8mm Illuminated Tactile Switch

- Surface mount gull-wing terminal and standard through hole configurations
- Compatible with IR reflow, and manual soldering
- Single color display during illumination
- Variety of LED color options
- Variety of cap styles, colors, and laser etching options



Ordering Information

Series	Terminal Type	Actuating Direction	Actuation Force	LED Color	LED Type	Cap Style	Cap Color	Laser Etching	Packaging																																																				
228A	M	V	A	R	F	A	MS	A	R																																																				
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Note: Caps supplied in bulk for surface mount terminal constructions and must be installed after IR reflow soldering.

Cap Style

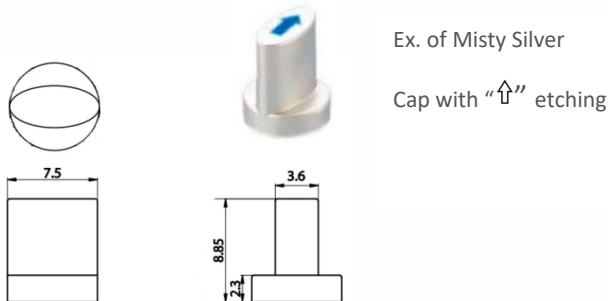
Code	Spec.
A	Elliptical column on round
B	45° Elliptical column on round
C	Cylinder on 3/4 circle

Cap Color

Code	Cap Color
MS	Misty Silver
W	White
BB	Bright Black
R	Red
MB	Misty Black
A	Orange
N	Natural

Cap Style Details

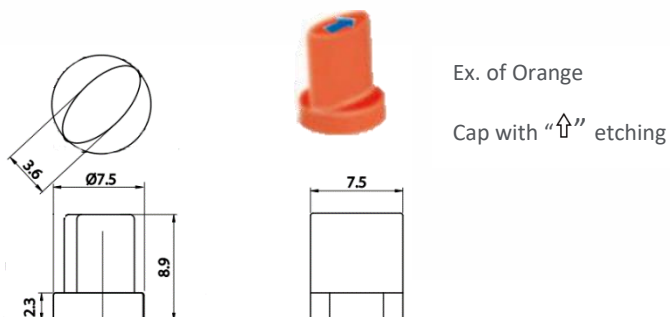
Style A: Elliptical column on round



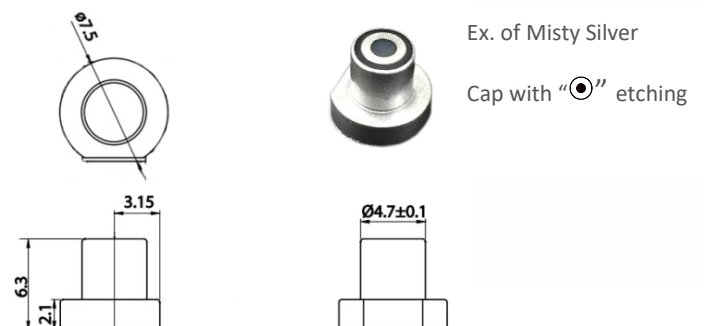
Cap Etching

Code	Style
A	↑
B	○
C	↑
D	OK
E	⊙

Style B: Elliptical column on round



Style C: Cylinder on 3/4 circle



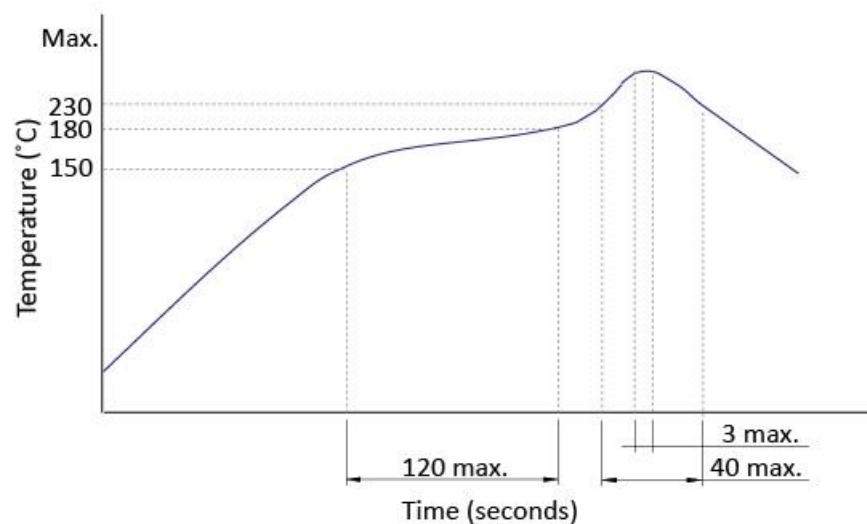
Electrical Specifications

Parameter	Conditions & Remarks	Min	Max	Unit
Contact Resistance			100	milliohms
Insulation Resistance	Between insulated terminals	100		megohms
Dielectric Strength	250 VAC between adjacent switches		1	minute
Nonswitching Rating			50 or 12	mA or VDC

Mechanical and Environmental

Soldering	Maximum reflow temperature, 260°C for 5 seconds Maximum manual temperature, 350°C for 3 seconds
RoHS	Lead-Free. Fully compliant to RoHS Directive 2011/65/EU
Operating Force	160±50 gf
Actuation Life	100,000 cycles
Actuator Strength	2 Kgf maximum
Travel	0.25±0.1 mm
Packaging	Standard tape and reel packaging for surface mount gull wing terminal Tray packaging for standard through hole
Operating Temperature Range	-25°C to +70°C
Storage Temperature Range	-30°C to +80°C

Soldering Profile



Mechanical Specifications

Figure 1 – 228ADV..., Standard Though Hole, Vertical Actuation

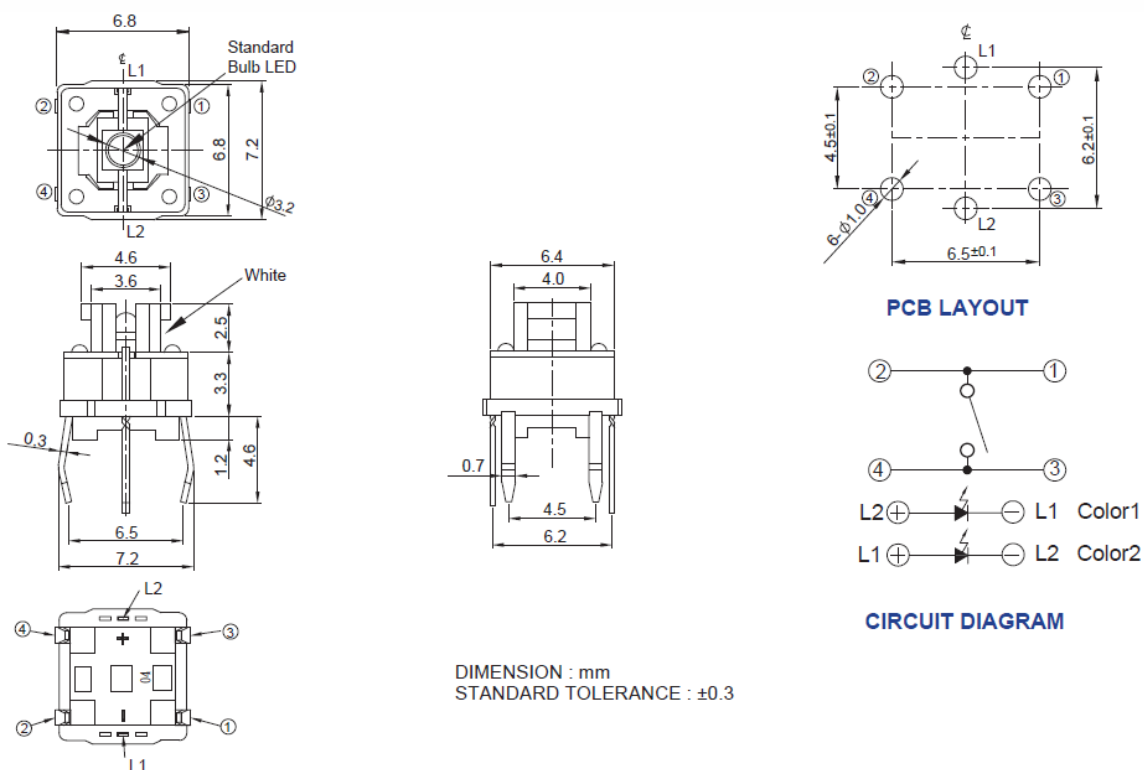


Figure 2 – 228AMV..., Surface Mount Gull Wing, Vertical Actuation

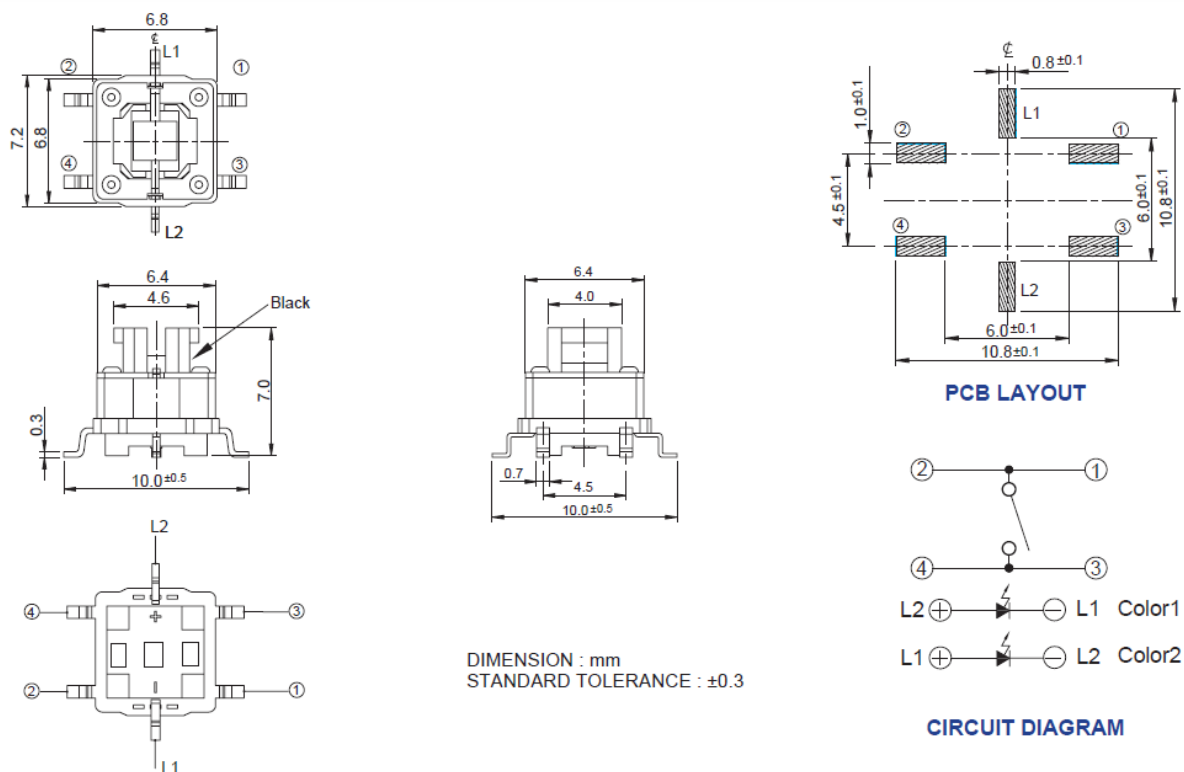


Figure 3 – 228ADS1..., Standard Though Hole, Side Actuation, 4 Pairs Terminals

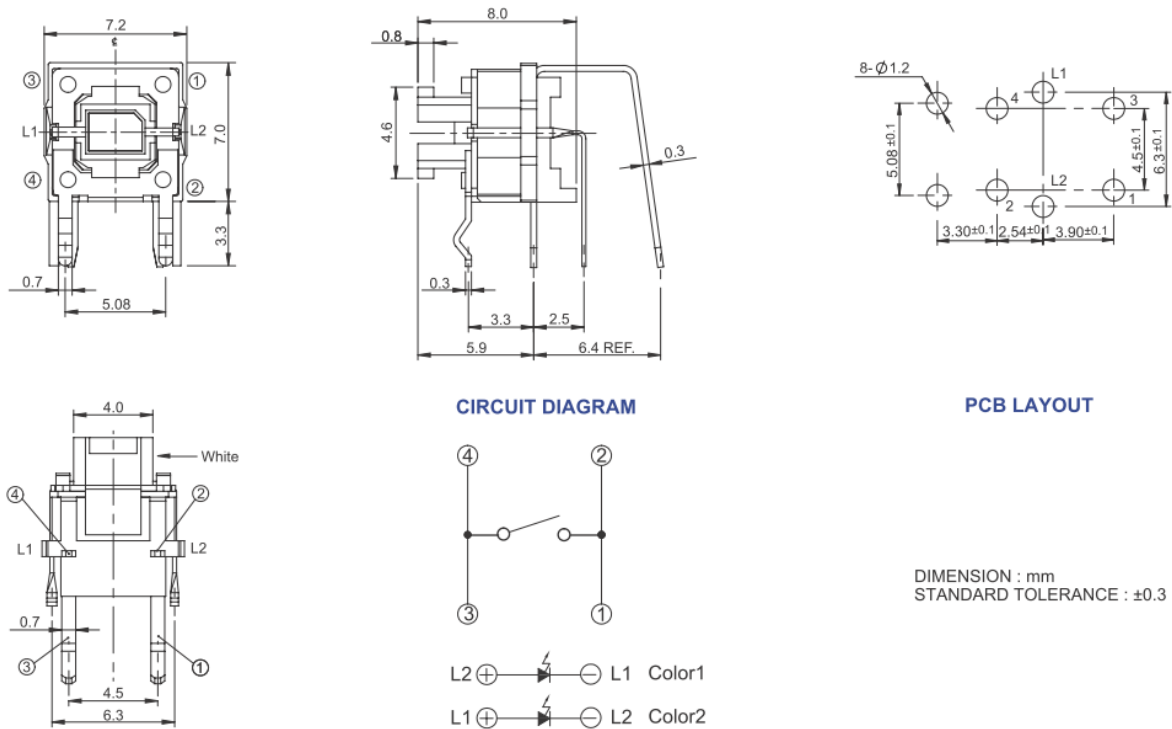


Figure 4 – 228ADVA...A..., Standard Though Hole, Vertical Actuation, Elliptical Column on Round Cap in Natural Color

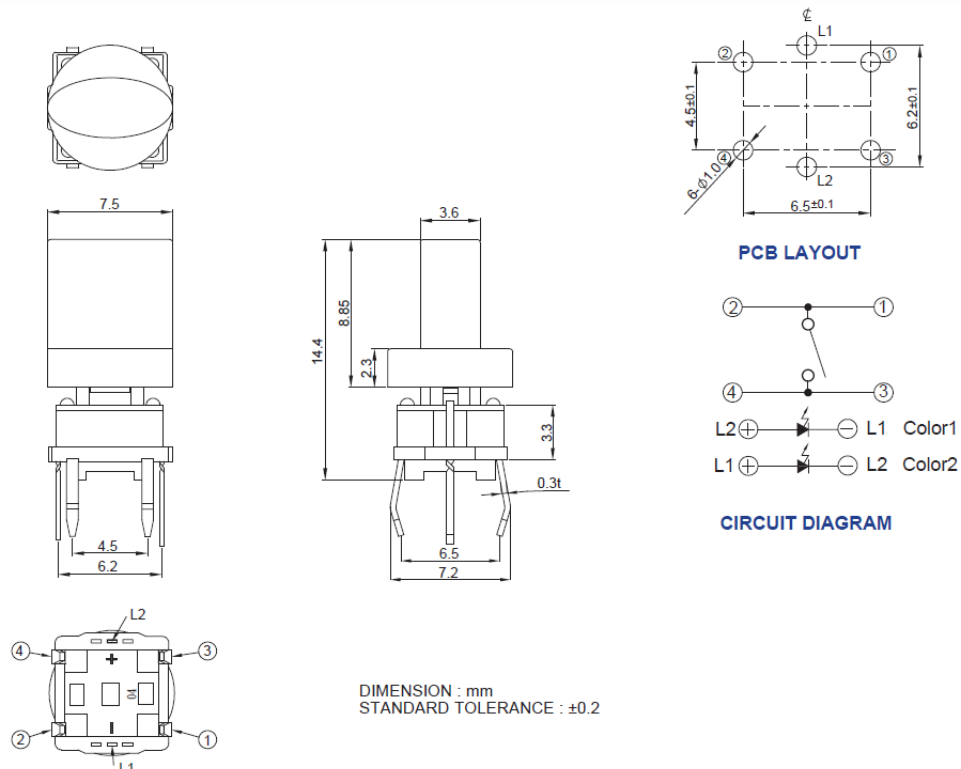
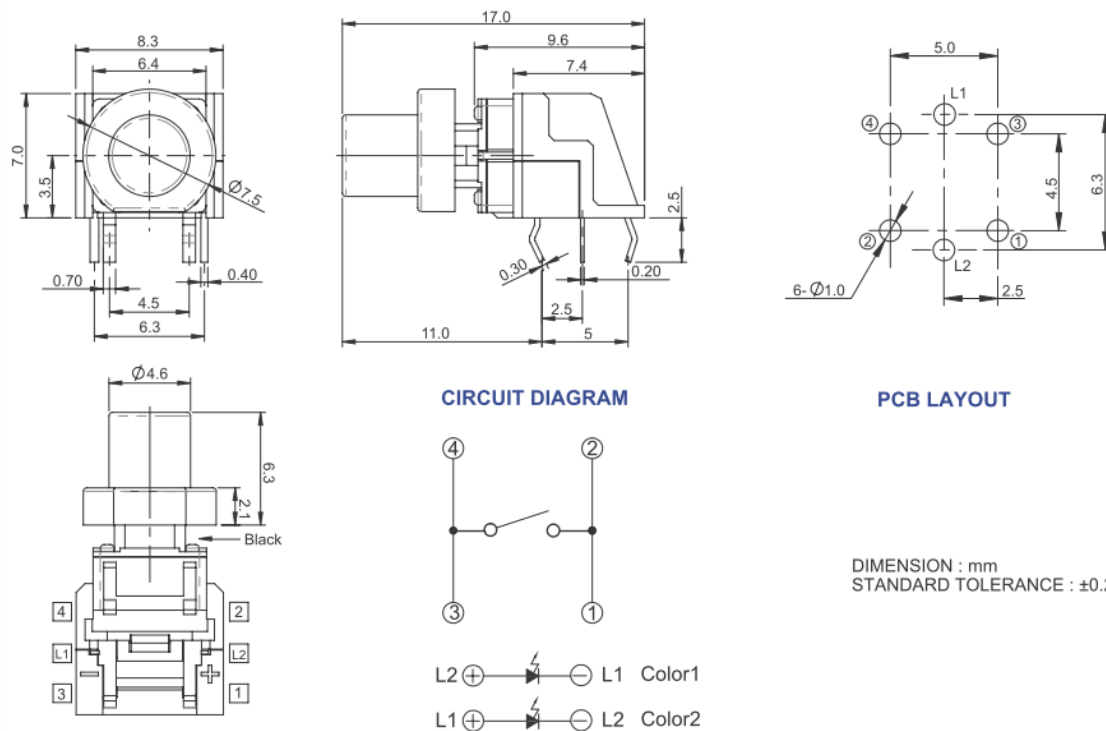


Figure 5 – 228ADS2...C..., Standard Though Hole, Side Actuation with 3 Pairs Terminals, & Cylinder on Round Cap

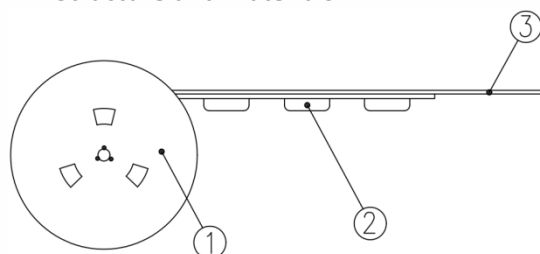


Packing: Tape and Reel

Surface Mount Gull Wing, Actuator Height 7.0mm

The packing specifications:

1. Structure and materials:



No	PART NAME	MATERIAL
①	REEL	POLYSTYRENE
②	CARRIER TAPE	POLYSTYRENE
③	COVER TAPE	POLYESTER

2. Packing quantity: 450 pcs/reel
3. More than 15 empty pockets should remain at each end of the carrier tape for each reel.
4. Shortage of less than 10 pcs a reel is acceptable but no more than 3 consecutive empty pockets are allowed.
5. Stripping strength of cover tape is between 10 gf to 130 gf and stripping angle should be within 165°~180°.
6. The product in the pocket of carrier tape should be placed in a specified correct position.
7. Tape and reel per EIA-481.
8. Dimensions:

